ChE422-Biochemical Engineering–Spring 2019
Student Learning Outcome 2
Rubric for Assignment 3
Protein engineering

SLO 1 Performance Indicator (PI)	**(1)** Unsatisfactory	**(2)** Developing	**(3)** Satisfactory
6. Solutions creativity alternatives	Demonstrates creative synthesis of solution and creates new alternatives by combining knowledge and information (< 6 pts)	Demonstrates solution with integration of diverse concepts or derivation of useful relationships involving ideas covered in course concepts; however, no alternative solutions are generated (6-7 pts)	Demonstrates creative synthesis of solution and creates new alternatives by combining knowledge and information (8-10 pts)

Averages:

Performance Indicators: (6) 3   Overall Average: 3

Average scores were fairly uniform, all students were strong in formulating the engineering system and model equations, and a majority 8/8 were satisfactory or higher in using appropriate resources needed to solve problems.

Biochemical Engineering Project Due on April 18

Design the experiments to express the following protein at the production scale. When you select an expression system, you need to consider the cost, yield, and feasibility (size, glycosylation and disulfide bond) of using the selected expression system. Your report should include

1. The (putative) function of the protein
2. DNA sequence of the protein:
    a. adjust the codons to the selected expression host.
    b. include the enzyme cutting site and other component for expression in the sequence
3. Molecular cloning procedures
    a. describe the key steps for the cloning, selection, and verification (gene and protein)
    b. provide detailed information about expression vector and cloning site
    c. provide the sequences of both cloning primer and sequencing primer
4. Host cells and expression conditions

MNLQAQPKAQNKRKRCLFGGQEPAPKEQPPPLQPPQQSIRVKEEQYLGHEGPGG
AVSTSQPVELPPPSSLALLNSVVYGPERTSAAMLSQQVASVKWPNSVMAPGRGP
ERGGGGGVSDSSWQQQPGQPPPHSTWNCHSLSLYSATKGSPHPGVGVPTYYNH
PEALKREKAGGPQLDRYVRPMMPQKVQLEVGRPQAPLNSFHAAKKPPNQSLPL
QPFQLAFGHQVNRQVFRQGPPPPNPVAAFPPQKQQQQQPQQQQQQQAALPQ
MPLFENFYSMPQQPSQQPQDFGLQPAGPLGQSHLAHHSMAPYPFPPNPDMNPEL
RKALLQDSAPQPALPQVQIPFPRRSRRLSKEGILPPSALDGAGTQPGQEATGNLFL
HHWPLQQPPPGSLGQPHPEALGFPLELRESQLLPDGERLAPNGREREAPAMGSEE
GMRAVSTGDCGQVLRGGVIQSTRRRRRASQEANLLTLAQKAVELASLQNAKDG
SGSEEKRKSVLASTTKCGVEFSEPSLATKRAREDSGMVPLIIPVSVPVRTVDPTEA
AQAGGLDEDGKGPEQNPAEHKPSVIVTRRRSTRIPGTDAQAQAEDMNVKLEGEP
SVRKPKQRPRPEPLIIPTKAGTFIAPPVYSNITPYQSHLRSPVRLADHPSERSFELPP
YTPPPILSPVREGSGLYFNAIISTSTIPAPPPITPKSAHRTLLRTNSAEVTPPVLSVMG
EATPVSIEPRINVGSRFQAEIPLMRDRALAAADPHKADLVWQPWEDLESSREKQR
QVEDLLTAACSSIFPGAGTNQELALHCLHESRGDILETLNKLLLKKPLRPHNHPLA
TYHYTGSDQWKMAERKLFNKGIAIYKKDFFLVQKLIQTKTVAQCVEFYYTYKK
QVKIGRNGTLTFGDVDTSDEKSAQEEVEVDIKTSQKFPRVPLPRRESPSEERLEPK
REVKEPRKEGEEEVPEIQEKEEQEEGRERSRRAAAVKATQTLQANESASDILILRS
HESNAPGSAGGQASEKPREGTGKSRRALPFSEKKKKTETFSKTQNQENTFPCKKC
GR